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MILITARY SPECIFICATION

WELDING MACHINES, ARC, DC, GENERATOR, CONSTANT CURRENT, INTERNAL COMBUSTION ENGINE DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope: This specification covers direct current, (DC) constant current, engine driven welding machines with auxiliary power capability.

1.2 Classification. The welding machines shall be of the following types and sizes. The type and size machine required shall be as specified (see 6.2.1).

Type I - Air Cooled Gasoline
Engine Driven
Size 200 - 200 Amperes

Type II - Liquid Cooled Gasoline
Engine Driven
Size 200 - 200 Amperes
Size 250 - 250 Amperes
size 300 - 300 Amperes
Size 400 - 400 Anlperes
Size 600 - 600 Amperes
Size 650 - 650 Amperes

Type III - Liquid Cooled
Diesel Engine Driven
Size - 250 - 250 Amperes
Size - 300 - 300 Amperes
Size - 350 - 350 Amperes
Size - 400 - 400 Amperes
Size - 600 - 600 Amperes

Type IV - Air Cooled Diesel
Engine Driven
Size - 200 - 200 Amperes
Size - 400 - 400 Amperes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3431

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless othewise indicated, the issue in effect on the date of' invitation for bids or request for proposal-shall apply.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910)

Title 29 Code of Federal Regulations, Chapter XVII, Part 1910, and Amendments	Occupational Safety and Health Administration Standards

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM -380-79 - Metric Practice

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURING ASSOCIATION (NEMA)

EW-1-Electrical Arc-Welding Apparatus

WC-3 - Rubber-insulated wire and cable for the transmission and distribution of electrical engergy.

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(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, NW, Washington, DC 20037.)

3. REQIREMENTS

3.1 First article. When specified (see 6.2.1) and 6.3, the supplier shall furnish one complete welding machine of each type and size specified in the contract or order for first article inspection in accordance with Section 4.

3.2 Design. The welding machines shall be of the manufacturers standard commercial product and shall include any additional features necessary to comply with the requirements specified herein. Additional features which are not specified herein that are a part of the manufacturers standard product shall be included in the equipment furnished. A standard commercial product is a product which has been sold on the commercial market through advertisement, manufacturer's catalogs or brochures, and represents the manufactures latest production models. The welding machines shall consist basically of an internal combustion engine, a welding generator, engine and welding controls, a base frame and housing to enclose the entire machine. The machine shall be designed and rated to conform with the requirements of NEMA Standard EW-1, class I machines.

3.2.1 Safety and health requirements. Covers or other safety devices shall be provided for all parts of the machine presenting safety hazards. The covers and guards shall prevent inadvertent access to hazardous parts and shall be easily removable for maintenance and servicing. The machine shall comply with the requirements promulgated under OSHA that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.l and 6.4).

3.2.2 Interchangeability All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or the welding machine.

3.2.3 Materials. All material used in the manufacture of the machines shall be of the quality to produce machines to meet the requirements described herein.

3.2.4 Reclaimed materials. The welding machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.5 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the machine. When only one system. of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). in this specification, all measurements, dimensions,

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sizes and capacities are given in the U.S. Customary System of Units (US). These measurements may be converted to the International System of Units (SI) through the use of the conversion factors and methods specified In ASTM-E-380.

3.2.6 Lubrication. All bearings [except sealed-for-life type) and moving parts shall be provided with means for lubrication. All grease and oil fittings shall be easily accessible, without removal of other parts or components, and have means of sealing the openings to prevent dirt and foreign matter from entering.

3.2.7 Lifting eye. The machines shall be Provided with a lifting eye for handling by a crane or hoist. The eye shall be located at the top of the machine at the normal center of balance. The lifting eye shall be capable of supporting two and one-half times the weight of the machine and shall be mounted in such a manner that the machine and its accessories retain alignment during lifting operations.

3.2.8 Enclosure. The machine shall be contained in a weather-proofed enclosure or be inherently weatherproof. The Type II,III and IV machines shall have hinged or removable panels for access and operation. Design of the enclosure shall be such that internal parts are protected from the weather even when the enclosure is opened in the operating position, The sheet metal of the enclosure shall be not less than 0.045 inches in thickness.

3.2.9 Operating temperature. The welding generators shall conform to the operating temperature requirements of NEMA Standard EW-l.

3.2.10 Stabilit The machine shall be designed to operate satisfactorily when incline at an angle of 15 degrees in any direction from the normal operating position.

3.3 Construction. The machines shall be constructed so that when installed and operated, they can be used for any purpose for which they were designed. The machines shall be constructed of parts which are sound in structure and free from any repair.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. In no event shall such processes as welding, peening, plugging, or filling with solder or pastes be used on castings or forgings to reclaim any defective parts for use on the machines.

3.3.2 Welding, brazing, or soldering. Welding, brazing. or soldering shall be employed only where these operations are required in the original design.

3.3.3 Fastening devices. All screws, pins, bolts, and similar fasteners shall be Installed with means for preventing loss to tightness. fasteners subject to removal or adjustment shall not be swagged, peened, staked, or otherwise permanently deformed.

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3.3.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth. and all edges shall be either rounded or beveled unless sharpness is required to perform a necessary function.

3.3.5 Painting Unless otherwise specified (see 6.2.1), the welding machines shall be cleaned , treated, primed and painted in accordance with the manufacturer's standard commercial practice. Painting shall be a minimum of one primer coat and two finished coats. If a dry powdered epoxy process is used, a primer coat is not required.

3.3.6 Threads. All threaded parts shall be in accordance with FED-STD-H28.

3.4 Performance. The output ratings (amperes and volts) and performance capabilities of each machine shall be in accordance with NEMA-STD-EW-1 for class I machines. Unless otherwise specified, each machine shall be rated at not less than 60% duty cycle. When specified (see 6.2.1) size 350 and size 650 machines shall be rated at 100% duty cycle. The machines shall provide a satisfactory power source for producing the welds specified herein, for producing auxiliary electric power specified herein, and for the following welding and related operation:

a. Manual welding with all types of DC covered electrodes of all sizes within the machine's capacity, and in all Positions for which the electrodes are suitable.

b. Automatic stud welding of steel studs up to 1/8 inch for size 200 ampere machines, up to 1/4 inch for size 300 ampere machines, up to 3/8 inch for size 400 ampere machine, and up to 7/16 inch for size 600 and 650 ampere machines at 20 percent duty cycle. When a stud welding gun is required under accessories, the bend test specified in 4.6.2.3 shall be conducted.

c. Air carbon-arc cutting at amperage equal to the machine's rated output at rated duty cycle.

3.5 Components,

3.5.1 Engine and accessories.

3.5.1.1 Engine. The engines for the welding machine shall be standard commercial internal combustion engines of current design and development. The engines shall develop horsepower to operate the welding generator, with accessories, at its maximum rated load. The engine for the Type I machine shall be a 4-cycle, forced air cooled gasoline engine which will operate satisfactorily on a regular grade of gasoline. The engine for Type II machine shall be a 4-cycle, liquid cooled gasoline engine which will operate satisfactorily on a regular grade of gasoline. The engines for the Type 111 and

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Type IV, machines shall be a 2 or 4 cycle diesel which will operate satisfactorily on #2 grade diesel fuel. The engine for the Type 111 machine shall be liquid cooled and the engine for the Type IV machine shall be forced air cooled. All engines shall comply with government emission standards.

3.5.1.2 Starting system. Unless otherwise specified the Type I machine shall be manually started by a hand crank or automatic recoil pull cord. When specified (see 6.2.1) the Type I machine shall have a 12 volt electrical starting system. The Type 11 and III, and IV machines shall have a 12 volt electrical starting system. Electrically started machines shall have adequate cranking capacity to start the engine under all conditions specified herein. Unless otherwise specified (see 6.2.1) , required batteries shall be provided with each machine and the batteries shall be of the maintenance-free type or dry charged (Electrolyte is not required).

3.5.1.3 Engine lubrication system. Each engine shall have a pressure type lubrication system with Provisions to drain the oil from the crankcase. The grade and capicity of the oil supply shall be specified on the filler cap or adjacent surface. When specified (see 6.2.1) the engine shall be provided with a low oil pressure shut-down device to stop the engine in the event of inadequate oil pressure.

3.5.1.4 Choke controls. The Type I and II machines shall have a choke control system to aid in starting the engine.

3.5.1.5 Battery charging generator. When a 12-volt starting system is required, the machines shall have a battery charging generator or alternator and regulator of sufficient size and capacity to keep the battery charge for starting the engine.

3.5.1.6 Governor. The governor shall maintain the speed of the engine within five percent of the operating speeds required for the performance of the machine. The Type III and IV machines shall have an overspeed protective device that will stop the engine if maximum R.P.M. is exceeded.

3.5.1.7 Starting aid. The Type III and IV machine shall have a cold weather starting device. It shall be the volatile fluid type designed to inject the fluid through a connection in the air intake manifold: A warning statement as to when the starting aid fluid shall, and shall not be injected shall be printed on a corrosion resistant plate and secured near the starting aid control.

3.5.1.8 Air cleaner. The engines shall have an air cleaner to filter all air for the air intake system. The air cleaner shall be of the wet or dry type, design for easy disassembly and cleaning.

3.5.1.9 Oil filter. Each machine shall have a replaceable cartridge type oil filter.

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3.5.1.10 Cooling system. Liquid cooled engines shall be provided with thermostat controlled liquid cooling system of adequate size and design to operate at an ambient temperature between -25°F and 110°F. Unless otherwise specified (see 6.2.1) liquid cooled engines shall have a high temperature shutdown device to stop the engine in the event of excessive temperature.

3.5.1.11 Automatic idling device. Each machine shall have an automatic idling device to reduce speed of the engine to an idle when not weldlng. Upon striking an arc, the idling device-shall immediately allw the engine to speed up to its previous setting. The idling device shall have the means to keep the engine at a constant speed if so desired. The idling device shall have a time delay to allow the operator when welding, to change electrode before the engine comes to idling speed.

3.5.1.12 Muffler. The machine shall have a muffler on the engine exhaust system. If the system is the vertical or upright type, it shall have a suitable rain cap on the end of the exhaust pipe to prevent water from entering.

3.5.1.13 Controls and instruments. The engine controls and instruments shall be grouped together in a separate panel, or be integral with the welding generator control panel. The controls for the cold weather starting aid may be located on the engine. Liquid cooled engines shall have a temperature gage or warning light, each engine shall have an oil pressure gage or warning light and engines with an electrical charging system shall have an ameter or warning light, All instruments shall be flush mounted, When specified (see 6.2.1) the engine shall have an hour meter of the sealed repeating type to register up to 9999.9 hour.

3.5.2 Fuel system,

3.5.2.1 Fuel tank. When operated at maximum load and rated duty cycle, the Type I machine shall have a fuel capacity to operate not less than 3 hours, the Type II machine not less than 6 hours and the Type III and IV machines not less than 8 hours. Each tank shall have a drain plug.

3.5.2.2 Fuel gage. The Type II, III and IV machines shall have a gage to register the quantity of fuel in the tank.

3.5.2.3 Fuel lines. Fuel lines shall be installed free of sharp bends and loops that might cause failure from fatigue, strain, or vapor lock. The fuel lines shall be bracketed and fastened when necessary to prevent vibration. A shut-off cock shall be installed in the outlet of the tank when the tank is located to provide gravity feed.

3.5.2.4 Fuel strainer and filter. The Type I and II machines shall have a strainer in fuel system. The strainer shall have a removable cover or bowl for cleaning and servicing. The Type III and IV machine shall have means

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to filter the fuel before it enters the fuel injection pump. The filter shall be of the replaceable cartridge type.

3.5.3 Welding generator, exciter, and auxiliary power supply.

3.5.3.1 Welding generator. The generator shall conform to the standards in the NEMA Publication EW-1. The generator shall be the conventional constant current self-regulating type, with a drooping volt-amoere characteristic. The rotor shall have a double-sealed anti-friction bearing, and shall be close coupled to the engine to maintain alignment with the engine flywheel.

3.5.3.2 Exciter. The exciter shall be integral with the welding generator, or be separate and mounted on the same shaft as the main welding generator. The exciter shall have fast voltage buildup, and be designed to prevent polarity reversal. The commutator, brushes and holders shall be readily accessible for servicing and maintenance.

3.5.3.3 Insulation. The maximum allowable temperature rise of insulated parts of the machine shall not exceed the temperature limits of NEMA-EW-1 for the class of insulation system used.

3.5.3.4 Generator instruments. The generator for the Type II, III and IV machines shall have a voltmeter to register the voltage at the welding output terminal and an ammeter to register the current during the welding operation. A combination volt-ammeter may be used. The meters shall be flush mounted and adequate to within three percent of full scale reading.

3.5.3.5 Generator output terminals. The terminals shall be of the stud, and wing or hex nut type. The studs shall be not less than 1/2 inch in diameter and shall be recessed-, protected, or located to prevent accidental contact. For units not equipped with a polarity reversing switch, the output terminals shall be identified as positive or negative.

3.5.3.6 Generator controls. The controls shall be mounted in a panel on the welding generator end of the machine. The controls shall be designed for easy operation with a gloved or mittened hand.

3.5.3.6.1 Current control. Two controls shall be provided to control and set the desired welding current range. One control or series of taps shall control the current through the entire range of the machine; the other shall be provided for fine stepless adjustments within the settings of the current control.

3.5.3.6.2 Remote current control. When specified the Type II, III and IV machines shall have a remote control to make fine adjustments at the work on the coarse generator current control. The remote control cable shall have connections suitable for using longer extensions. The length of the remote control cable shall be as specified see 6.2.1).

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3.5.3.6.3 Polarity switch. For the Type II, III and IV machines, a positive acting switch shall be provided to change the polarity of the welding current. The switch shall have its position marked for either straight or reverse polarity.

3.5.3.7 Auxiliary electric power supply. The machine shall have an auxiliary power supply.A NEMA Standard 115 volt duplex receptacle shall be provided for lights and power tools. The circuit shall be overload protected by a circuit breaker or fuses. When specified a 230 volt AC auxiliary power supply shall be provided in addition to the 115 volt circuit. The 230 volt circuit shall have a NEMA Standard through wire single or duplex receptacle. The KW capacity of the 230 volt circuit shall be as specified (see 6.2.1).

3.5.4 Wiring The type and size of wire used shall have sufficient capacity, temperature an voltage rating for the intended use. Wiring subject to chafing or vibration shall be secured to prevent damage. The wire and terminals shall be color or number coded. A wiring diagram of the generator and engine shall be provided and fastened in a convenient location to the inside surface of the machine enclosure. The diagram shall have a transparent covering or protective coating.

3.5.5 Control and instrument marking. The controls and instruments shall be legibly marked for identification and position. The instruments may be marked and identified on the face of the instruments.

3.5.6 Machine mounting Unless otherwise specified, the welding machine shall be furnished with a standard base,arranged and drilled for bolting the machine to the floor or deck. When required, the machine shall be mounted on skids or mounted on a trailer for portability in a shop or work area as specified (see 6.2.1).

3.5.6.1 Over the road trailer. When specified (see 6.2.1), the machine shall be mounted on a two-wheel or four-wheel tandem trailer with tube or tubeless tires, not less than 600 x 14 size. The tires shall be standard pneumatic commercial type having a ply rating and size recommended in the tire and rim association year book for the gross weight of the unit. The trailer shall be designed in such a manner that, when fully loaded, it may be towed at speeds up to 50 mph on improved roads and up to 10 mph on rough terrain. When being towed at any speed up to 50 mph, the trailer shall track-true with minimum side sway or hunting and shall support the complete welding machine and accessories. Road clearance to the trailer's lowest menber shall not be less than eight inches. The trailer shall have resilient mounting (springs, torsion bars, or similar devices) to minimize transmission of road shock to the welding machine. When specified (see 6.2.1) a spare tire and wheel shall be provided and mounted in an accessible area on the trailer.

3.5.6.1.1 Provisions for accessories. Space shall be provided on over the road trailers for firmly mounting two gas cylinders and storage of all hoses and welding accessories-. Gas cylinder-mounting assembly(s) shall be provided to accommodate one gas cylinder up to 9-1/4 diameter and one gas cylinder up to

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12-7/8 inch diameter. The mounting assembly shall be positioned so as not to interfere with the control panel, hinder daily servicing, or add to the design volumn when the gas cylinders are installed. The installed gas cylinders shall not affect the fore and aft balance of tandem axle units.

3.5.6.1.2 Drawbar. The drawbar tongue shall be approximately 48 inches long and shall have a lunette type ring fabricated of one inch round steel. The ring shall have an inside nominal diameter of three inches.

3.5.6.1.3 Lights reflectors and chains. Vehicle reflectors,lights and safety chains shall be installed on the trailer in accordance with the manufacturer's standard practice but shall comply with pertinent ICC regulations for vehicles to be towed on highways.

3.5.6.1.4 Parking brake. The trailer shall be equipped with a parking brake sufficient to hold the fully loaded trailer on a 15 degree incline.

3,5.6.1.5 Landing gear. Two wheel trailers shall be equipped with a retractable or adjustable type landing gear with full swiveling bottom wheel. A height adjustment of not less than four inches above level to four inches below level position shall be provided.

3.6 Equipment.

3.6.1 Standard equipment. The machine shall be furnished with all standard equipment normally furnished by the manufacturer.

3.6.2 Accessories. The following accessories shall be furnished:

One set of welding cables. The cables shall be fitted with an electrode holder, a ground clamp, and two cable lugs that will fit the terminals on the machine. The length and size of the cables shall be as s ecified (see 6.2.1, Table I and 6.6). "Unless otherwise specified (see 6.2.1), cables shall conform to class M cable of NEMA Standard WC3 part 7 including part 2 and part 6. The size of the cable shall be selected from the Table for Copper Conductors, class M of NEMA WC3 nominal size, suggested construction and nominal strands, and from the Table for ampacities of portable cables. Cables shall be compatible with the maximum ampere rating and duty cycle of the welder concerned. Unless otherwise specified (see 6.2.1)A the jacket shall be natural rubber, or styrene butadiene rubber (SBR) (140° to -35°F) or neopene (140° F to -35°F). or nitrile-butadiene polyvinyl chloride compound (for -35 F) or polyethylene black (167 to -67 F) conforming to NEMA WC3 requirements for heavy-duty jacket for portable arc welding cable.The standard single lead is considered to be 50 feet long (100 foot both leads combined). Table I is provided as a guide for welding cable current-carrying capacity and voltage drop. The Table (see Table I), is valid for up to 60 percent duty cycle. When specified (see 6.2.1), a 10 to 12 foot whip lead shall be provided for connecting to the electrode holder and the welding lead and quick detachable (QD) twist lock couplings for connecting the whip lead to the welding lead and each 50 foot section of the ground and welding leads. The

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TABLE I.

WELDING CABLE CURRENT-CARRYING CAPACITY AND VOLTAGE DROPS

Based on Copper at 140°F and 104°F Ambient Temperature												
MAXIMUM LENGTHS												
	100 FEET		150 FEET		200 FEET		250 FEET		300 FEET		400 FEET	
CABLE SIZE	Current Capacity in Amperes	DC Voltage Drop (volts)	Current Capacity Amperes	DC Voltage Drop	Current Capacity Amperes	DC Voltage Drop	Current Capacity Amperes	DC Voltage Drop	Current Capacity Amperes	DC Voltage Drop	Current Capacity Amperes	DC Voltage Drop
3	150	3.70	100	3.70	75	3.70	75	4.62	----	----	----	----
2	200	3.92	150	4.40	125	4.90	100	4.90	75	4.41	50	3.92
1	250	3.88	200	4.63	150	4.64	125	4.85	100	4.65	75	4.65
1/0	300	3.72	250	4.65	200	4.96	150	4.68	125	4.65	100	4.96
2/0	375	3.68	300	4.40	250	4.88	200	4.90	150	4.41	125	4.90
3/0	450	3.51	375	4.38	300	4.68	225	4.38	200	4.68	150	4.68
4/0	550	3.41	450	4.17	375	4.65	250	3.88	250	4.60	200	4.96
250 MCM	600	3.12	---	----	---	----	---	----	---	----	---	----
300 MCM	650	2.80	---	----	---	----	---	----	---	----	---	----

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QD shall be Factory Mutual approved or meet OSHA standards. Unless othewise specified, insulators shall be water, oil, solvent, and ozone resistant. Cables shall be tested for resistance and cross sectional area (see 4.6.2.2).

b. Unless otherwise specified (see 6.2.1) provisions on the machine for storage of the welding cables, remote control unit and tools shall be provided.

c, When specified (see 6,2.1), a stud welding gun and connectors compatible with the welder(s) concerned shall be provided.

3.7 Suppression of radio interference. When required (see 6.2.1), radio interference suppression shall be as specified.

3.8 Nameplate. A corrosion resistant metal nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below:

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Engine make, model
Amperage output at rated volts
Auxiliary electric power KW, Voltage, AC or DC and frequency
Date of manufacture
Contract or order number
National Stock Number or Plant Equipment Code

3.9 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.10 Workmanship. Workmanship of the machine and accessories shall be commensurate with the requirements spectfied herein, and shall be of a quality prevailing among manufacturer's normally producing equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless othewise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follws:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the first article sample when submitted in accordance with 3.1. The first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the first article to pass the examination or one or more perproduction tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherw!se specified, quality conformance inspection shall consist of the examination in 4.5, the tests in 4.6,1, and the inspection in 4.7. When additional quality conformance tests are required, they shall be as specified (see 6.2.1) and selected from the test herein. Failure of the item to pass the examination or any test shall be cause for rejection.

4.5 Examination. The machine and equipment shall be examined for design, construction, materials, components, electrical equipment, and workmanship to determine compliance with the requirements of this specification,

4.6 Tests.

4.6.1 Operational test, The machine shall be operated to check the output in each range and to check the output of the auxiliary circuit along with the function of all instruments and controls. A resistive load shall be used to perform these tests.

4.6.2 Performance test. To demonstrate the capability of the machine to weld with its specified range, a weld shall be made on two 1/2 inch by four inch by six inch mild steel plates on straight polarity, then a similar weld on reverse polarity using AWSE-6011 welding rods. The finished weld shall be inspected by the methods specified by the procuring activity (see 6.2.1). Evidence of difficulty in striking or maintaining an arc during the welding or the inability to produce a sound weld shall constitute failure of test and rejection of welding machines.

4.6.2.1 Rain test. The machine shall be subjected to a weather proof test by spraying fresh water at a rate of 4-inches + l-inch per hour at an angle of 30 to 45 degrees from vertical. The test shall be performed with water directed against all four sides and the roof of the machine simultaneously and continuously for not less than one hour. Immediately follwing the water spray the machine shall be examined for water penetration and damage and the machine shall start and be operated to meet the performance requirements specified herein,

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4.6.2.2 Lead cable test. When specified (see 6.2,1), lead cables shall be tested as described in NEMA WC-3 to determine conformance as indicated below. A Certificate of Conformance from an independent laboratory acceptable to the Government may be accepted in lieu of testing.

(a) Direct current resistance, Tested by method for direct current resistance of conductors. Resistance shall not exceed the parameters for portable arc welding copper cables.

(b) Cross sectional area. Tested by method for cross sectional area for stranded conductors. The cross sectional area shall be not less than 98 percent of the area given in the Table.

4.6.2.3 stud welding bend test. When a stud welding gun is required with the machine the gun shall be tested to demonstrate stud welding capabilities. Two studs of each diameter (1/8, 1/4, 3/8 or 7/16 inch, depending on welder involved see 3.4.b) from cold drawn bar stock of 55,000 psi minimum, shall be stud welded to a base membier of thickness equal to 1/2 stud diameter and two studs each of applicable diameter shall be stud welded to a base member with thickness equal the stud diameter. Each welded stud shall be bent (by striking with a hammer) to an angle of 30 from their original axis. Any cracking or breaking in the weld shall be cause for rejection.

4.6.3 Emission test. When specified (see 6.2.1), the engine shall be run at idling speed and no load and the exhaust tested to determine compliance with emission requirements. The test shall then be repeated with the engine at full load and operating speed. Exhaust emissions shall be within limits allowed by Government regulations.

4.6.4 Parking brake test. The fully loaded trailer shall be placed on a 15 degree incline and the parking brake applied. The engine of the welding machine shall be started and run at normal operating speed. There shall be no movement of the trailer down the incline under the above conditions.

4.6.5 Trailer road test. When a over the road type trailer is required it shall be tested for roadability. The fully loaded trailer shall be towed for 10 miles at approximately 50 mph on improved highways and 10 miles at approximately 10 mph on rough terrain. During this test, the trailer shall track steady, with minimum side sway, and cause no difficulty in handling the towing vehicle. At the end of this test, the welding machine shall be examined for damage due to road shock. Failure to track steady, difficulty in handling the towing vehicle, or damage due to road shock shall be cause for rejection.

4.7 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

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5. PACKAGING

5.1 Preservation, packing and marking. Unless otherwise specified (see 6.2.1), preservation, packing and marking shall conform to the requirements of MIL-STD-1188.

6. NOTES

6.1 Intended use. The machines covered by this specification are intended to provide a DC power source for welding where it is not practical to use electric power or It is not available, where portability is desired in shops, field maintenance and construction, and to provide auxiliary electric power as desired in connection with the welding operation.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Type and size required (see 1.2).
c. First article, when required (see 3.1).
d. Additional safety and health requirements (see 3.2.1).
e. Measurement system, if different (see 3.2.5).
f. Painting, if different (see 3.3.5).
g. Duty cycle if different (see 3.4).
h. Electric starting system, (Type I machine) if required (see 3.5.1.2).
i. Specify when batteries are not required (see 3.5.1.2).
j. Low oil pressure safety shut down, if required (see 3.5.1.3).
k. Specify if excessive temperature shutdown device is not required on liquid cooled engine (see 3.5.1.10).
l. Hour meter if required (see 3.5.1.13).
m. Remote current control and cable length, if required (see 3.5.3.6.2).
n. Auxiliary 230 volt power supply, if required, specify KW capacity (see 3.5.3.7).
o. Machine mounting, if different (see 3.5.6).
p. Over the road trailer mounting if required (see 3.5.6.1).
q. Spare tire and wheel, if required (see 3.5.6.1).
r. Specify length and size of welding cables (see 3.6.2a).
s. Specify if welding cables not conforming to NEMA Standard WC3 are acceptable (see 3.6.2a).
t. Welding cable jacket material, if different (see 3.6.2a).
u. Welding cable whip leads, when required (see 3.6.2a).
v. Specify if provisions for storage of cables and accessories on the machine are not required (see 3.6.2b).
w. Stud welding gun when required (see 3.6.2c).
x. Suppression of radio interference, if required (see 3.7).

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y" Quality conformance inspection if different (see 4,4).
z, Test weld inspection method (see 4.6.2).
aa. Lead cable test, if required (see 4.6.2.2).
bb. Engine emission test, if required (see 4.6.3.).
cc. Preparation for delivery, if different (see 5.1).

6.2.2 Contract data requirements, Required technical data, such as operators manuals, parts lists, wiring diagrams, and other instructions for operation and maintenance as identified on a numbered DD Form 1664 should be specified on a DD Form 1423 incorporated into the contract.

6.3 First article. When a first article is required, it shall be tested and approved under the appropriated provisions of 7-104.55 of the Defense Acquisition Regulation. The first article should be a preproduction sample, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instructions regarding arrangements for examination, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.3 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with 'those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants in the the environment (and does not limit the hazard level of individual machine in the environment) the procuring agency is advised to analyze the existing hazards levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. If specific point-of-operation guarding is required, the procuring agency should specify the exact configuration of the guard required. The above, and any other additional safety and health requirements should be specified in detail under 6.2.ld.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.6 When selecting welding lead cable, it Is recommended that the voltage drop does not exceed 4 volts in the total lead circuit. When selecting leads for cold ambient temperatures the pliability of the jacket,when cold should be considered. A jacket suitable for arctic use should be selected even if the resistance to oil, solvents, and ozone must be sacrificed.

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Review Activities:
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Navy - SH, YD
Air Force - 99
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Using Activities:
Navy - MC, OS

Preparing Activity:
DLA- IP

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